



結合專家規則與上下文感知網路之高吞吐量情感分析 ——以航空業文字與表情符號為例

A High-Throughput Sentiment Analysis Framework Combining Expert Rules and
Context-Aware Networks for Aviation Text and Emojis

Presenter: Tsung-Lin Yang Advisor: Prof. Sok-Ian Sou



Outline

1. Introduction

2. Related Work

3. System Design

4. Experiment setting

5. Performance Evaluation

6. Conclusion

The Necessity of Emojis in Multimodal Sentiment Analysis



Sentiment analysis can be used to analysis :

- **customer perspectives**
- **service experiences**
- **brand reputation**

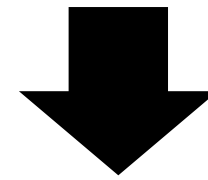
Emoji can be used to express more emotional nuance that text cannot capture.

The Core Challenge: Sarcasm & Polarity Conflict

Polarity Conflict :

The most disruptive issue in text-emoji combinations.

Ex : Positive Text + Negative Emoji $\neq$ Positive Sentiment



Distorted Predictions :

Failure to identify semantic contradictions leads to severe accuracy drops.

Misguided Strategies :

Results in incorrect service recovery and misallocated business resources.



The First Study to Incorporate Emojis in Airline Reviews

Baseline Study:

"Sentiment analysis of emoji fused reviews using machine learning and Bert"
Amit Khan et al.,(Nature 2025)

Key Contribution:

The first paper to integrate emojis into the sentiment analysis of airline reviews

Challenging:

1. The Performance Trade-off: Accuracy vs. Computational Cost
2. Lack of Decision Transparency

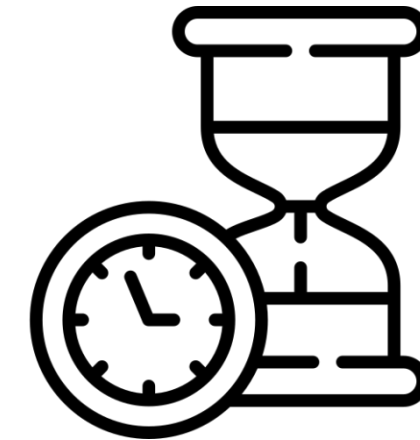
The Performance Trade-off: Accuracy vs. Computational Cost

Heavy Deep Learning Models

e.g. Cross-Attention Mechanisms



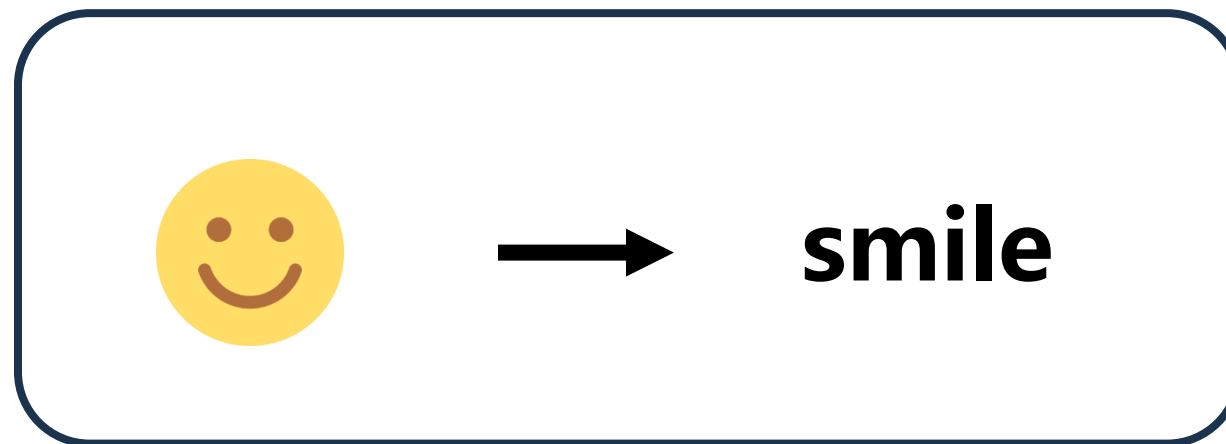
- High accuracy on static datasets



- Excessive Latency
- Cannot meet real-time monitoring speed

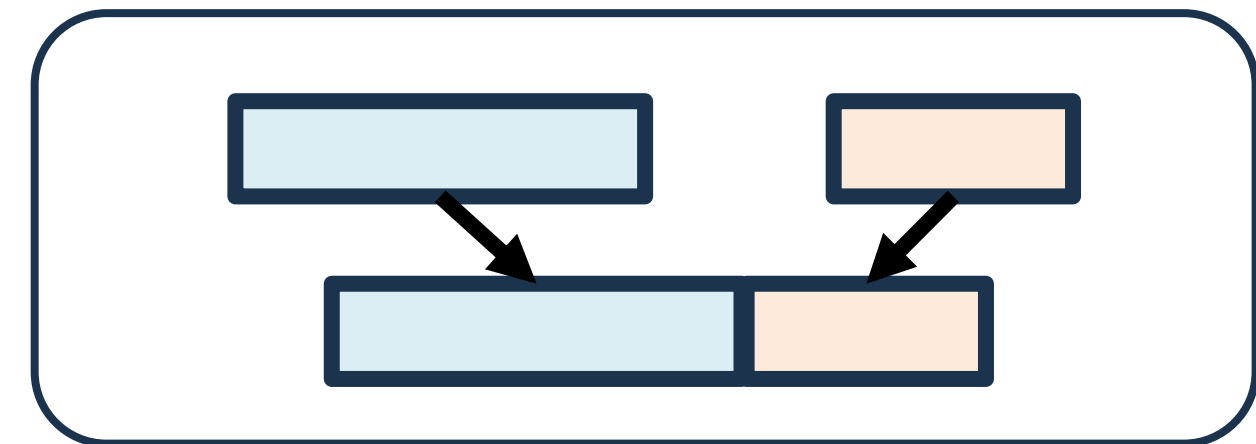
The Performance Trade-off: Accuracy vs. Computational Cost

Text Replacement



- Feature Erasure

Concatenation



- Lack of Semantic Interaction

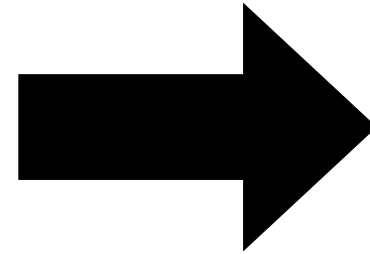


Sarcasm Detection

Lack of Decision Transparency



Black Box



Barrier to Commercial Adoption

Proposed Model: Hybrid Rule-based and CLN-DistilBERT Architecture

Contribution:

- **A Novel Two-Stage Multimodal Framework:**
Integrates Emoji2Vec with CLN to dynamically fuse text and emoji features.
Effectively resolving polarity conflicts, leading to enhanced sentiment classification accuracy.
- **High-Efficiency Inference via Early Interception:**
Employs a rule-driven early exit mechanism to bypass heavy computations, significantly reducing latency.
- **Built-in Decision Interpretability:**
Provides transparent, rule-based reasoning for extreme sarcasm prediction through front-end interception



Related Work



Evolution of Emoji Processing Strategies : Text Replacement

Replacing emojis with text descriptions

Ex: “Great service 😏” ➡ “Great service **annoyed**”

Limit:

- Models confuse converted emoji text with original text, ignoring the unique role of emojis in sentiment reversal and emotional emphasis.

➡ Result in misclassification in text-emoji semantic conflicts.



Evolution of Emoji Processing Strategies : Lexicon-Based Mapping

Assigns a 1D polarity score to emojis based on sentiment co-occurrence in massive corpora.

Advantages:

- Features faster querying speed and white-box transparency.

System Application:

- Combined with VADER to achieve faster interception of extreme semantic conflicts.

Limit:

- 1D scores fail to capture deep contextual nuances.



Evolution of Emoji Processing Strategies : Emoji2Vec

Maps emojis into a 300-dimensional continuous vector space.

Rich Semantic Nuance:

- Overcomes 1D limits by capturing deep, context-dependent implicit features.

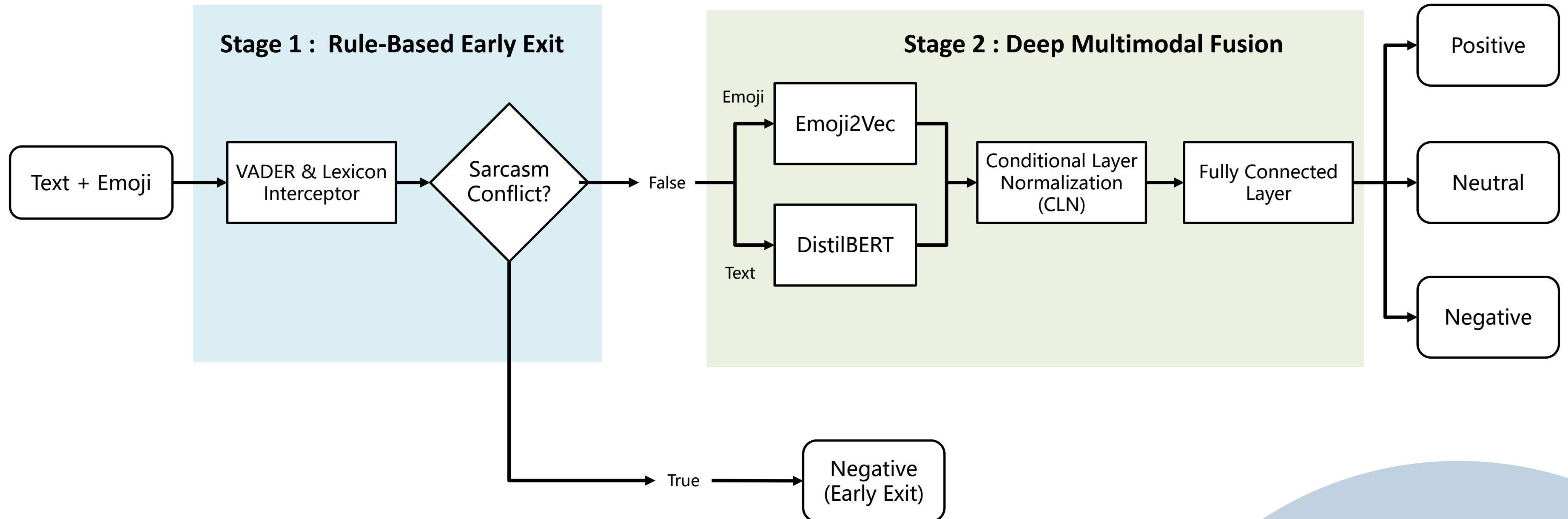
System Application:

- Serves as the foundational pre-trained embedding in the backend analysis layer.
- Integrates with the Conditional Layer Normalization (CLN) network for deep multimodal feature fusion with text hidden states.



System Design

System Architecture Overview





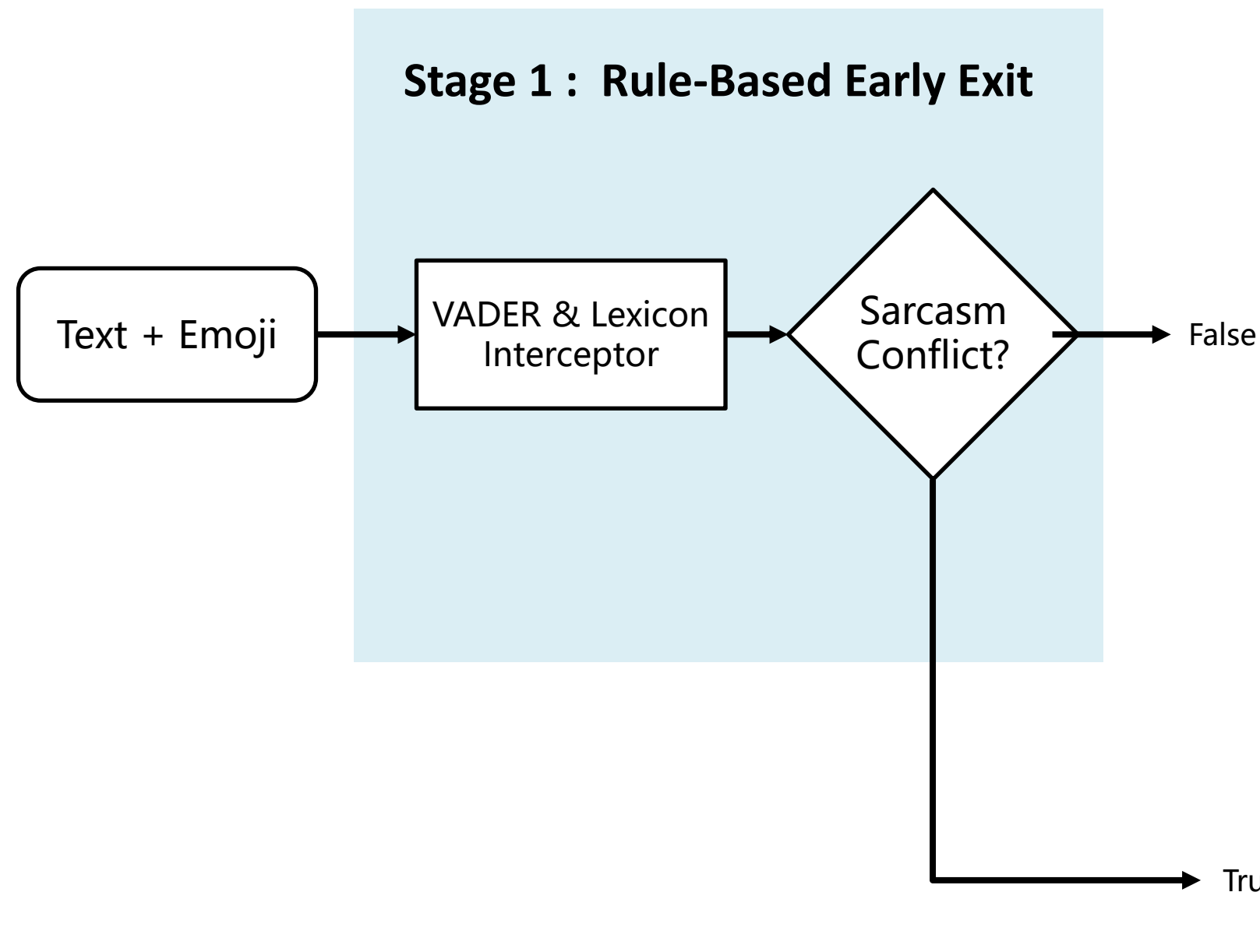
Notation List

S_{vader}	the text sentiment polarity score extracted by VADER
S_{emoji}	the net sentiment score of emojis
$C_{positive}$	the occurrence count of the emoji in positive texts across the corpus.
$C_{negative}$	the occurrence count of the emoji in negative texts across the corpus.
C_{emoji}	the total occurrence count of the emoji within the corpus.

Notation List

$\boldsymbol{x}$	The text feature vector
$\boldsymbol{y}$	The emoji feature vector
$E[x]$	The expected value (mean) of the text feature vector
$Var(x)$	The variance of the text feature vector
W_γ, W_β	The learnable weight matrices
ϵ	A small constant added for numerical stability
$\odot$	The element-wise multiplication

Stage 1: Ruled-Based Early Exit Module

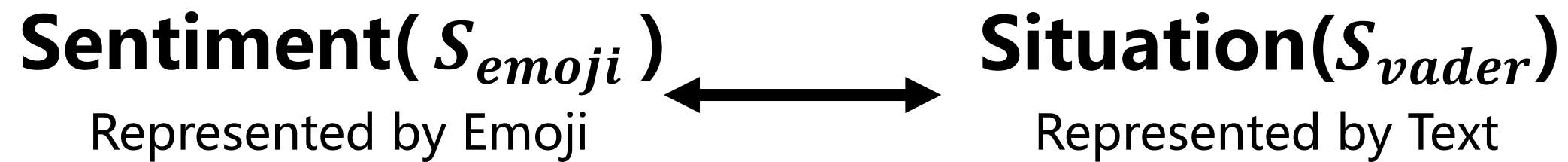


- **Target:**
Prevent excessive latency from heavy neural networks.
- **Mechanism:**
VADER & Lexicon polarity check.

VADER :

Human-annotated lexicons combined with semantic reasoning heuristics to calculate the sentiment score.

Ellen Riloff (ACL 2013) 《Sarcasm as Contrast between a Positive Sentiment and Negative Situation》



Pre-Condition : $C_{emoji} > 100$

$$S_{emoji} = \frac{C_{positive} - C_{negative}}{C_{emoji}}$$

If $(S_{emoji} > 0.5 \quad \text{AND} \quad S_{vader} < -0.4)$ OR
 $(S_{emoji} < -0.1 \quad \text{AND} \quad S_{vader} > 0.4)$

⇒ Trigger Early Exit

S_{vader} the text sentiment polarity score

$C_{positive}$ the occurrence count of an emoji labeled as positive

$C_{negative}$ the occurrence count of an emoji labeled as negative

C_{emoji} the total occurrence count of the emoji in emoji dictionary

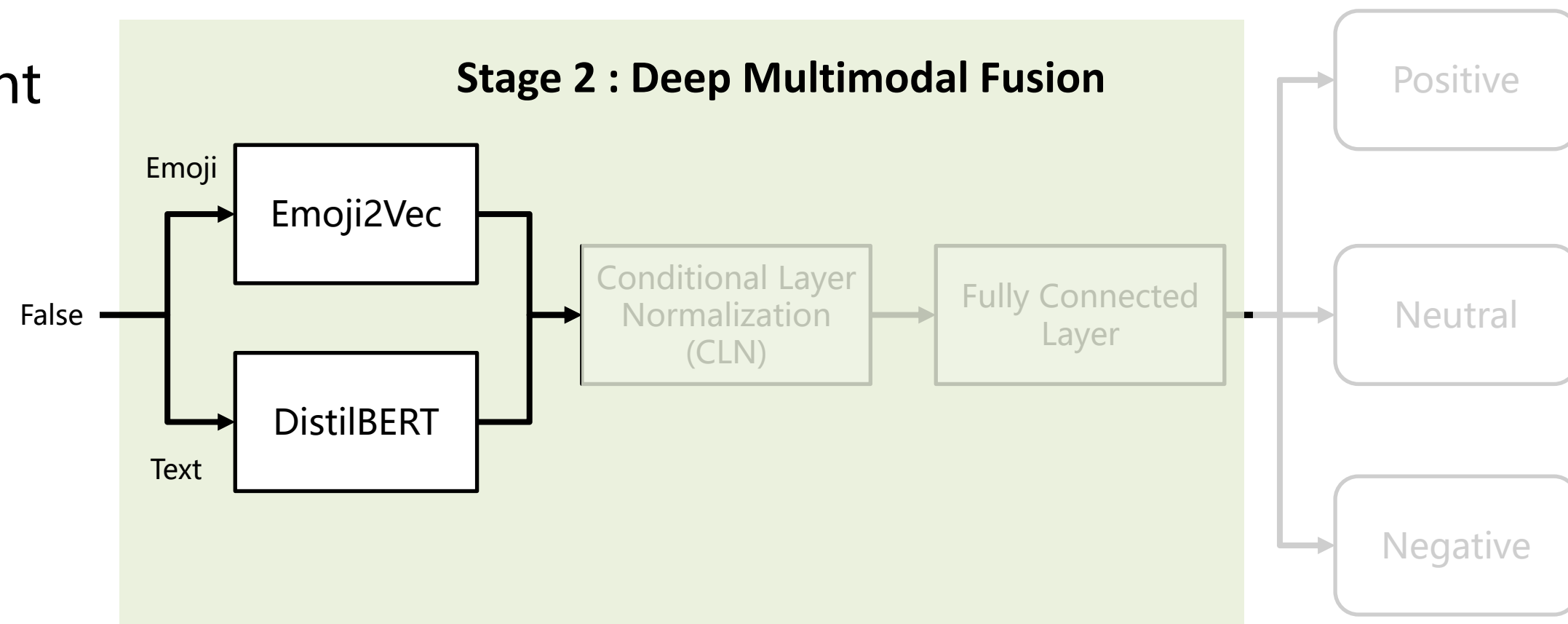
Stage 2: Deep Multimodal Fusion Module

Emoji2Vec : Emoji Representation

- Maps emojis to a 300D continuous latent space.

DistilBERT : Text Encoding

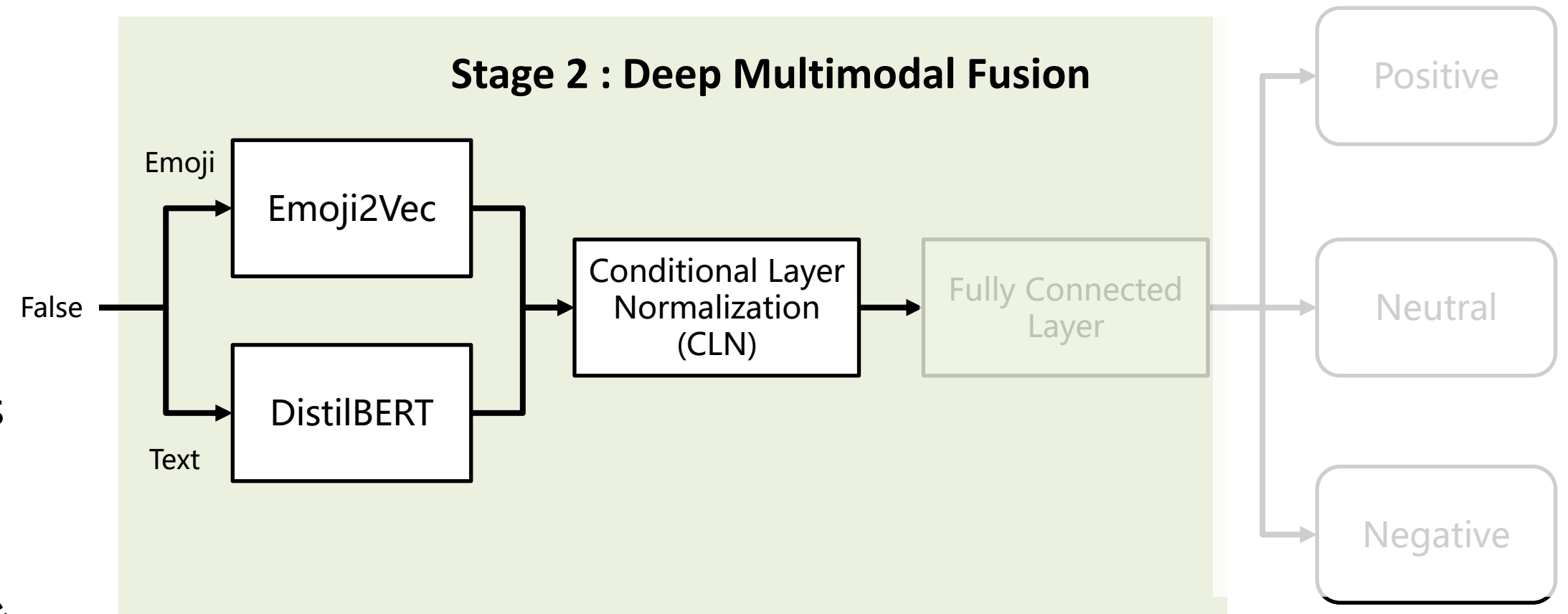
- Applies knowledge distillation for lightweight inference.
- Retains 97% language understanding capacity.
- Reduces 40% parameters for low-latency inference.



Stage 2: Deep Multimodal Fusion Module

Fusion: Conditional Layer Normalization (CLN)

- Models emojis as global modulators to capture implicit semantic reversal.
- Dynamically scales and shifts text features based on emoji conditions.



x The text feature vector

y The emoji feature vector

W_γ, W_β The learnable weight matrices

$$\text{CLN}(x, y) = \frac{x - E[x]}{\sqrt{\text{Var}[x] + \varepsilon}} \odot W_\gamma y + W_\beta y$$



Experiment Setting



Dataset Preparation & Data Augmentation

The original data (Khan et al.) exhibits significant **class imbalance**
(negative : 9169 , neutral : 3082 , positive :2354)

We introduce the **Back Translation** technique to minority classes (neutral & negative)
Retains polarity while increasing syntactic diversity. (negative, neutral, positive: 9169)

English $\longrightarrow$ French $\longrightarrow$ English

Data Split : Train 80% , Validation 10% , Test 10%

Data Preprocessing

Tokenization

- Splitting sentences into the smallest meaningful units
- "I like it" → ["I", "like", "it"]

Lemmatization (LEM)

- Normalizing words to their base form
- "Finalized", "Finally" → "Final"



Data Preprocessing

Stop Word Removal (RSW)

- Eliminating common words that carry little contextual meaning
- e.g a, an, the, about, over

Removal of digits and special symbols (SYM)

- Removing digits and punctuation marks



Backbone Model Selection Experiment

Goal:

Evaluating the **computational cost** versus **classification accuracy** of various lightweight language models on social media texts.

Methodology:

Integrating different base models (BERT, ALBERT, DistilBERT, ELECTRA) with the Conditional Layer Normalization (CLN) feature fusion layer.



Preprocessing Ablation Study

Goal:

To investigate how traditional NLP preprocessing affects the model's understanding of extreme emotions and semantic conflicts.

Methodology:

- Established a Full Pipeline (LEM + RSW + SYM) as the baseline.
- Conducted an ablation by independently turning off each module (w/o LEM, w/o RSW, w/o SYM).



Dual-Scenario Ablation Study

Objective of Dual-Scenario Study

- Verify commercial robustness & defense capability.
- Dual-Scenario
 - Standard Scenario
 - Stress-Test Scenario

Why Extreme Scenario Testing?

- Real-world Threat: "Black Swan" events (e.g., IT outages, mass strikes)
- Sudden influx of extreme, semantically complex sarcasm.



Dual-Scenario Ablation Study

Synthesized 500 extreme sarcasm samples via LLM (Gemini 3.1 Pro).

Topic includes :

1. Digital Booking & Ticketing
2. Ground Services & Boarding
3. Flight Schedule & Logistics
4. Cabin Hardware & In-Flight Experience
5. Baggage Handling & Post-Flight

Dual-Scenario Ablation Study

Explicit cross-modal polarity conflicts

- **Type A: Positive Text + Negative Emoji**

Ex : Thank you for this beautiful and magnificent demonstration of entirely stationary luggage processing today! 🙄

- **Type B: Negative Text + Positive Emoji**

Ex : Horrible missing flight attendants caused an awful abysmal delay ruining my terrible stressful trip completely. 🎉



Parameter

GPU : NVIDIA RTX A5000

Batch Size : 16

Optimizer : AdamW

Learning Rate : 3×10^{-5}

Epochs : 10

Max Length : 128

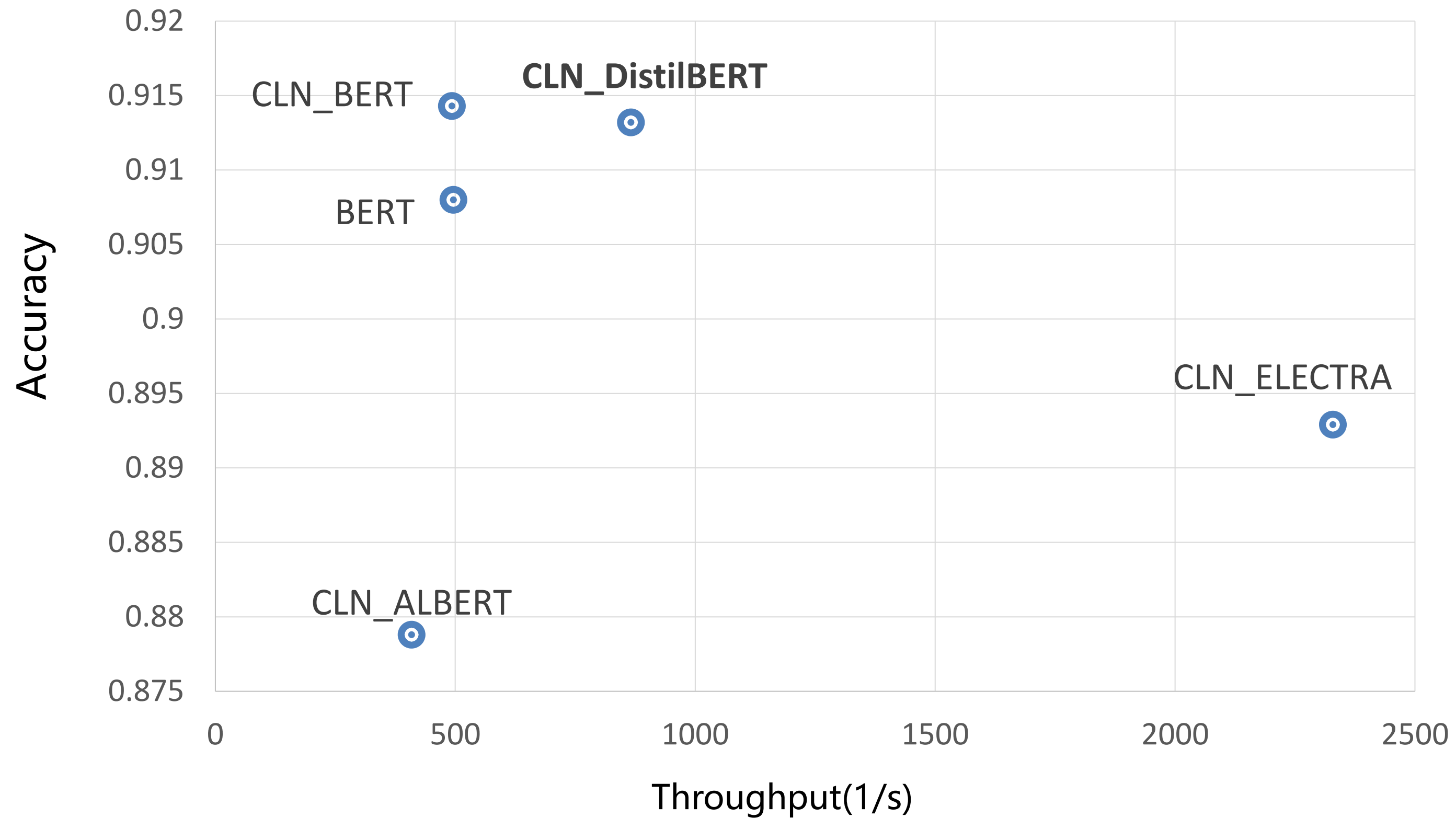


Performance Evaluation

Backbone Model Selection: Accuracy vs. Throughput

	Accuracy	Throughput
BERT(base)	0.908	496
CLN_BERT	0.914	493
CLN_ALBERT	0.879	409
CLN_DistilBERT	0.910	866
CLN_ELECTRA	0.893	2329

Performance Trade-off Analysis



Ablation Study: The Superiority of Raw Text

Ablation Setting	Accuracy	Throughput
CLN_DistilBERT(Full)	0.910	918
CLN_DistilBERT w/o LEM	0.910	893
CLN_DistilBERT w/o RSW	0.931	885
CLN_DistilBERT w/o SYM	0.928	911
CLN_DistilBERT(Raw)	0.938	911

LEM: Normalizing words to their base form

RSW: Eliminating common words that carry little contextual meaning

SYM: Removing digits and punctuation marks

Contextual Disruption:

Removing stop words fragments the semantic continuity

Feature Erasure:

Stripping punctuation and symbols destroys critical markers used to express extreme emotions.

* Consistent with empirical findings: "Is text preprocessing still worth the time?" (Siino et al., 2024, ScienceDirect).

Ablation Study on Robustness: Effectiveness of Two-Stage Architecture

	Standard Scenario		Stress-Test Scenario	
	Accuracy	Throughput	Accuracy	Throughput
BERT(base)	0.934	498	0.868	471
CLN_BERT	0.938	471	0.869	462
CLN_DistilBERT	0.938	911	0.868	903
CLN_DistilBERT_(with stage1)	0.935	911	0.917	942



Conclusion

Contributions

- **A Novel Two-Stage Multimodal Framework:**
Integrates Emoji2Vec with CLN to dynamically fuse text and emoji features, enhancing sentiment classification accuracy.
- **High-Efficiency Inference via Early Interception:**
Employs a rule-driven early exit mechanism to bypass heavy computations, significantly reducing latency.
- **Built-in Decision Interpretability:**
Provides transparent, rule-based reasoning for extreme sarcasm prediction through front-end interception

Limitations

- Emoji semantics evolve over time in social media

Future Work

- Implement dynamic updating mechanisms for the static lexicon.



Thank
You